

Industrial A5S Sensor with AOC(4-20mA)-AOV(0-10v) Datasheet

1. Product Overview

WIN-SN-A5S-AOC-AOV is a versatile data acquisition device designed for real-time monitoring in industrial applications, especially ducts. It features precise sensors as detailed, and provides analog output options of **10-bit and 12-bit resolution**, ranging from **0 to 10V or 4-20mA**. The board's low power consumption and wide operating temperature range make it suitable for numerous applications, such as **outdoor air quality monitoring, climate control in greenhouses, and environmental monitoring systems**.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-A5S-AOC-AOV ensures **accurate Sensor (CH₄/H₂S/CO₂/CO/NH₃/PH₃/C₂H₄) data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IOT applications**.



Ex: - Product Image

WIN-SN- A5S –AOC-AOV

Sensor Parameters

	SN	Sensor Name	Measuring Parameter	Measuring Range	Output	Operating Temperature
☐	a	A-5S-CO	CO	0-1000 ppm	AOC/AOV	-20°C to +50°C
☐	b	A-5S-CO2	CO2	0-5000/ 10000/ 40000 ppm	AOC/AOV	-20°C to +50°C
☐	c	A-5S-NH3	NH3	0-100 ppm	AOC/AOV	-20°C to +50°C
☐	d	A-5S-PH3	PH3	0-1000 ppm	AOC/AOV	-20°C to +50°C
☐	e	A-5S-CH4	CH4	0-10000/ 20000 ppm	AOC/AOV	-20°C to +50°C
☐	f	A-5S-H2S	H2S	0-100 ppm	AOC/AOV	-20°C to +50°C
☐	g	A-5S-C2H4	C2H4	0-20 ppm	AOC/AOV	-20°C to +50°C

	Product Configuration
☒	A5S+AO+C
☐	A5S+AO+V

2. Technical Parameters

a) CO Sensor

- Target Gas: Carbon Monoxide(CO)
- Detection Range: 0 to 1000ppm
- Accuracy: ± 1 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

b) CO2 Sensor

- Target Gas: Carbon Dioxide (CO2)
- Detection Range: 0 to 5000/10000/40000 ppm
- Accuracy: ± 1 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$

- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

c) NH3 Sensor

- Target Gas: Ammonia (NH₃)
- Detection Range: 0 to 100ppm
- Accuracy: ± 1 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

d) PH3 Sensor

- Target Gas: Phosphine (PH₃)
- Detection Range: 0 to 1000ppm
- Accuracy: ± 1 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

e) CH4 Sensor

- Target Gas: Methane (CH₄)
- Detection Range: 0 to 10000/20000 ppm
- Accuracy: ± 50 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

f) H2S Sensor

- Target Gas: Hydrogen Sulfide (H₂S)
- Detection Range: 0 to 100 ppm
- Accuracy: ± 1 ppm or $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

g) C2H4 Sensor

- Target Gas: Ethylene (C2H4)
- Detection Range: 0 to 20 ppm
- Accuracy: Resolution of 0.5 ppm (Zero drift 0~1 ppm)
- Detection Accuracy: Repeatability < ±2% output signal
- Technology: 3 Electrodes Electrochemical
- Response Time: Less than 90 seconds

Note: All sensors require a 90-second warm-up period after power-on to provide accurate measurements.

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

5. Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

Recommended Cable Electrical Characteristics

- Read all instructions before using the sensor board
- Keep the sensor board away from water or moisture to prevent damage.
- Do not disassemble or modify the sensor board.
- Use only the specified power sources and mentioned.

6. Analog Output AOC/AOV Specifications

- **Measured Parameters:** A5S sensor.
- **Output Type:** Analog Output-10-bit and 12-bit resolution.
- **Output Range:** AOV 0-10V / AOC 4-20mA.
- **Wide temperature range:** –20 °C to +85 °C.
- **Full humidity range:** 0 % to 100 % RH
- **Product dimension:** 85*85*40 mm.
- **Power supply DC:** 12-24V DC @1A (Type –C power USB).
- **Certifications:**



Contact Information

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